Flowers trimerous.
Monocolpate (1 furrow) Pollen is present.

Anatomical features

- Vascular bundles are scattered in the stem
- Vascular bundles are closed (Cambium absent).
- Secondary growth is absent.

Current Angiosperm Phylogeny Group (APG) System of classification doesn't recognize dicots as a monophyletic group. Plants that are traditionally classified under dicots are dispersed in several clades such as early Magnolids and Eudicots.

Summary

Plant Kingdom includes Algae, Bryophytes, Pteridophytes, Gymnosperms and Angiosperms

The life cycle in plants fall under three types 1. Haplontic,2. Diplontic and 3. Haplodiplontic

Algae are autotrophic, chlorophyll bearing organisms. The Plant body is not differentiated into root like, stem like or leaf like structures. A wide range of thallus organization is found in algae. They reproduce vegetatively through fragmentation, tuber and akinete formation. Zoospores, autospores and hypnospores are produced during asexual reproduction and sexual reproduction occurs through isogamy, anisogamy and oogamy.

Chara is a fresh water alga and is popularly called "Stone Worts". The plant body is multicellular, macroscopic and is differentiated into main axis and rhizoids. Sexual reproduction is oogamous.

Bryophytes are the simplest land plants. They are called amphibians of plant kingdom or nonvascular cryptogams. The plant body is gametophyte. The sporophyte depends upon gametophyte. Conducting tissues like xylem and phloem is absent. Vegetative reproduction takes place through fragmentation, formation of adventitious bud and gemmae. Sexual reproduction is oogamous. Water is essential for fertilization.

Marchantia belongs to the class Hepaticopsida. The thallus is dorsiventral and is attached to the substratum by means of rhizoids. The internal structure of the thallus reveals the presence of photosynthetic region and a storage region. Vegetative reproduction takes place through fragmentation and formation of gemmae. The sexual reproduction is oogamous. Sporophyte bears spores. Alternation of generation is present.

Pteridophytes are also called vascular cryptogams. The plant body is sporophyte and is long lived, which is differentiated into root, stem and leaves. They may be homosporous or heterosporous. The sporangia with spores are found in sporophylls. The sporophylls organise to form cones or strobilus. The spores germinates to produce haploid, multicellular heart shaped independent gametophyte called prothallus. Sexual reproduction is oogamous. The life cycle shows alternation of generation.

The term stele includes central cylinder of vascular tissues comprising xylem, phloem, pericycle, endodermis and pith . There are two major types of stele namely protostele and siphonostele.

Selaginella belongs to the class Lycopsida. The plant body is sporophyte. It is differentiated into stem, leaf, rhizophore and roots. Heterospory is found and two types of spores namely microspores and megaspores are produced in sporangia. The microsporangia and megasporania are borne on sporophylls. The sporophylls are organized to form cone. Sexual reproduction is oogamous. Alternation of generation is present.

Gymnosperms are naked seed producing plants. The plant body is sporophyte and it is the dominant phase. Coralloid roots are found in *Cycas*. The roots of *Pinus* possess

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